

Utkarsh Pandey

+91 88308 45957 | utkarsh16103@gmail.com | linkedin.com/in/utkarsh-pandey-0392482a6 | github.com/ShreyPan

SUMMARY

Backend developer with hands-on experience building production-grade REST APIs in Spring Boot (Java) and Node.js, with security exposure from API penetration testing at Tata Communications. Skilled in JWT auth, Docker, MongoDB, and MVC architecture; currently building AdaptIQ, a full-stack fitness platform with adaptive analytics.

EDUCATION

Vellore Institute of Technology (VIT)

B.Tech – CSE with Specialization in AI & Robotics

Chennai, Tamil Nadu

Sep. 2022 – Present (Expected 2026)

JK Jr. College of Science and Arts

Senior Secondary (Science)

Pune, Maharashtra

Apr. 2020 – Jul. 2021

EXPERIENCE

AI/LLM-Powered Penetration Testing Intern

May. 2025 – Jul. 2025

Tata Communications

- Conducted manual and AI-assisted penetration testing on 10+ internal REST APIs and web applications, identifying critical authentication bypass and injection vulnerabilities.
- Integrated GPT-based LLM tools to automate payload generation, enhancing security test coverage and reducing manual effort by ~30% across regression test cycles.
- Intercepted and analysed HTTP traffic using Burp Suite across multiple test environments, documenting findings and recommended mitigations for the security team.

PROJECTS

AdaptIQ – Adaptive Fitness Platform (Full-Stack) | Java 21, Spring Boot, React.js, JUnit 5, SwaggerFeb. 2026 - Present

- Built 15+ REST endpoints (onboarding, program management, workout logging, session history, analytics) with avg. response time ~200ms; integrated with a React (Vite) frontend.
- Documented all APIs with Swagger/OpenAPI (springdoc), exposing interactive Swagger UI for faster integration and manual testing.
- Secured APIs with stateless JWT authentication, BCrypt hashing, CORS, and fine-grained Spring Security rules; implemented global exception handling via `@ControllerAdvice` with structured error responses.
- Designed Controller → Service → Repository architecture with DTOs and input validation; authored 20 unit tests (~85% service-layer coverage) using JUnit 5 and Mockito.
- Implemented adaptive analytics engine for fatigue scoring, estimated 1RM trend tracking, plateau detection, and weekly program adjustment recommendations.
- Containerised backend and MongoDB with Docker multi-stage builds and Compose; CI pipeline via GitHub Actions completes in under 2 minutes.

Expense Tracker | React, Node.js, Express, MongoDB, JWT, Tailwind CSS, GitHub Actions | Live Demo Jul. 2025

- Built and deployed a production-ready MERN expense tracker with React (Vite) frontend and Node.js/Express REST API backend, supporting income/expense CRUD, balance summaries, dashboard analytics, and user-specific financial records.
- Secured all endpoints with JWT authentication, BCrypt password hashing, protected routes, CORS, and layered server-side validation to prevent unauthorized access across all transaction workflows.
- Designed modular MongoDB schemas with per-user data isolation and role-based access control, enforcing strict cross-account data separation using MVC architecture.
- Implemented CI/CD pipeline with GitHub Actions for automated unit testing and Render deploy hooks, enabling reliable and repeatable production deployments.

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, Python, SQL, HTML/CSS

Backend: Spring Boot, Spring Security, Spring Data JPA, Node.js, REST APIs, JWT, Hibernate, BCrypt, Docker

Frontend: React.js, Vite, Tailwind CSS, Recharts

Databases: MongoDB

Architecture: MVC, Repository-Service-Controller, DTO Pattern, Layered API Design

Tools: Git, GitHub, Postman, Swagger/OpenAPI, Gradle, Burp Suite, JUnit 5, Mockito, GitHub Actions